

DEVIA Local SQLite and Firebase Schema Reference

For Last-Mile-Delivery-App trips, rider accounts, recent destinations, and post-trip/deviation questionnaires.

Current implementation note: local SQLite currently stores rider accounts, rider-code registrations, and recent destinations. Recorded trips are saved through Firebase rides. Post-trip/deviation questionnaire answers are currently persisted locally in app state storage and should be mirrored into Firebase using the recommended structure below.

Current SQLite Tables

local_accounts

Field	Type	Key	Description
phone	TEXT	PK	Rider phone number.
rider_code	TEXT		Assigned 6-digit rider code.
full_name	TEXT		Preferred/full name.
gender	TEXT		Gender from onboarding.
age_range	TEXT		Age range from onboarding.
city	TEXT		City/local area.
years_experience	TEXT		Delivery/riding experience.
accepted_policies	INTEGER		1 if rider accepted consent/policies.
created_at	TEXT		ISO timestamp.

rider_code_registrations

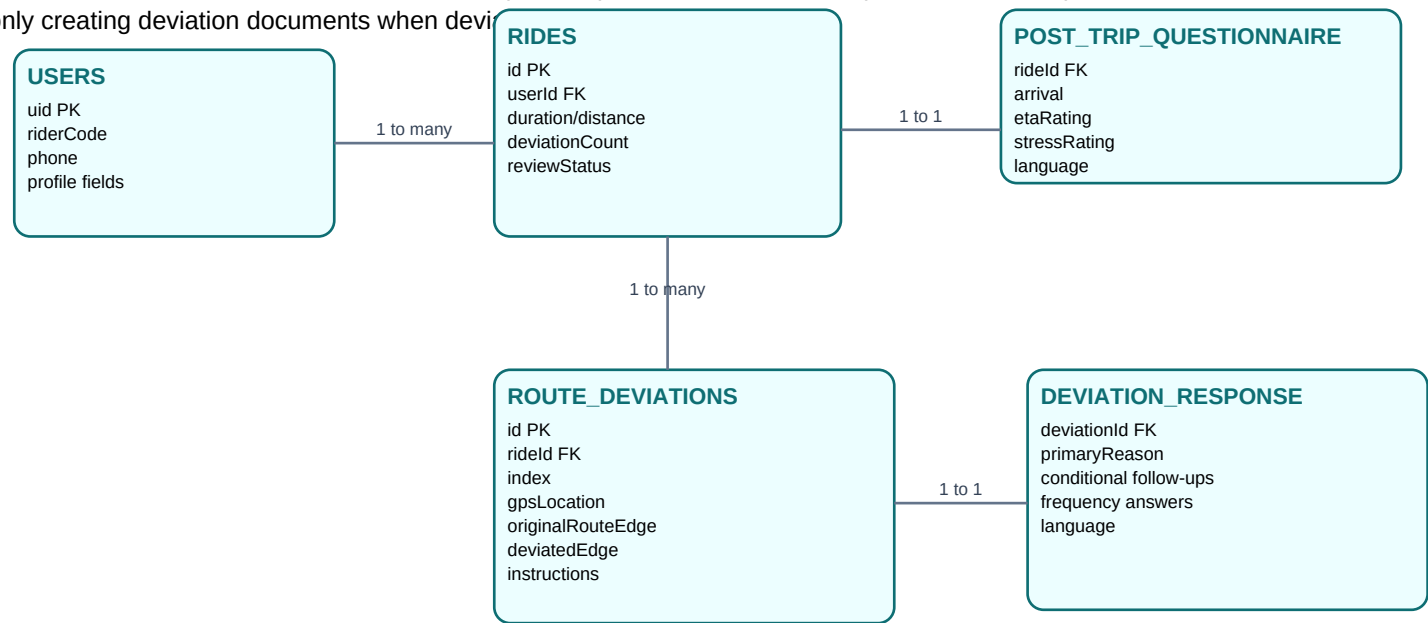
Field	Type	Key	Description
code	TEXT	PK	Valid 6-digit rider code. Includes temporary test code 123456.
phone	TEXT	FK-ish	Phone linked to local_accounts.phone.
registered_at	TEXT		ISO timestamp when code was registered.

recent_destinations

Field	Type	Key	Description
user_id	TEXT	PK part	Firebase UID or local user/account key.
name	TEXT	PK part	Destination display name.
full_address	TEXT		Full address from search provider.
longitude	REAL	PK part	Destination longitude.
latitude	REAL	PK part	Destination latitude.
updated_at	TEXT		Most recent selected/search timestamp.

Recommended Firebase ERD

Use this as the backend model for recorded trips and questionnaires. This keeps no-deviation trips valid and visible while only creating deviation documents when deviation occurs.



Firebase: users/{userId}

Field	Type	Description
uid	string	Firebase Auth UID.
riderCode	string	6-digit rider code.
phone	string	Phone used for compensation/contact.
fullName	string	Rider name.
gender	string	Onboarding detail.
ageRange	string	Onboarding detail.
city	string	Onboarding detail.
yearsExperience	string	Onboarding detail.
acceptedPolicies	boolean	Consent accepted.
createdAt	timestamp	Account creation timestamp.

Firebase: rides/{rideId}

Field	Type	Description
id	string	Ride document ID.
userId	string	Owner UID.
rideName	string	Trip name, e.g. New Delivery Trip.
startTime	timestamp/number	Trip start. Current app stores number ms.
endTime	timestamp/number	Trip end. Current app stores number ms.
duration	number	Recorded seconds.
distance	number	Recorded meters.
averageSpeed	number	Meters per second.
maxSpeed	number	Meters per second.
elevationGain	number	Meters.
deviationCount	number	0 when no deviations; greater than 0 when route deviations were detected.
reviewStatus	string	pending, reviewed, expired, invalid if needed.
createdAt	timestamp/number	Created timestamp.

Questionnaire Collections

Firestore: rides/{rideId}/postTripQuestionnaire/response

Field	Type	Description
rideId	string	Parent ride ID.
arrival	string	Canonical value: Early, On time, Late.
etaRating	number	1 to 5 rating.
stressRating	number	1 to 5 rating.
language	string	en or tl. Store canonical answers in English even if Tagalog labels are shown.
submittedAt	timestamp	Submission time.

Firestore: rides/{rideId}/deviations/{deviationId}

Field	Type	Description
id	string	Deviation document ID.
rideId	string	Parent ride ID.
index	number	0-based or 1-based deviation order. Be consistent.
originalRoute	string/null	Original route summary or polyline reference.
dateTime	timestamp	Deviation date/time.
isFaster	boolean/null	Whether deviation was faster if calculable.
hour	number	0-23 local hour.
dayOfWeek	string	Monday, Tuesday, etc.
dayType	string	Weekday or Weekend.
gpsLocation	GeoPoint/null	Deviation GPS point.
originalRouteEdge	string/null	Original route segment/edge.
deviatedEdge	string/null	Actual deviated segment/edge.
streetName	string/null	Street where deviation happened if available.
generatedInstruction	string/null	Expected instruction, e.g. Turn left.
deviationInstruction	string/null	Actual movement, e.g. Went straight.
imageUri	string/null	Map snapshot/image if implemented.

Firestore: rides/{rideId}/deviations/{deviationId}/response/current

Field	Type	Description
deviationId	string	Parent deviation ID.
rideId	string	Parent ride ID.
primaryReason	string	Canonical English reason.
primaryReasonOther	string/null	Typed answer when primaryReason is Other.
trafficSeverity	string/null	1 = Very Light through 5 = Severe.
rushHourCause	string/null	Yes, No, Unsure.
chooseDuringNonRush	string/null	Yes, No, Unsure.
blockageReason	string/null	Flood, Accident, Road closure, Illegal parking, Other.
blockageReasonOther	string/null	Typed answer when blockage reason is Other.
personalStopReason	array	Break, Meal, Restroom, Refuel, Took a Call, Accident/Repair, Other.
personalStopOther	string/null	Typed answer for personal stop Other.
stopDuration	string/null	< 5mins, 5-15mins, 15-30mins, > 30mins.
deviateAgainFrequency	string	Always, Often, Sometimes, Rarely, Never.
avoidRoadFrequency	string	Always, Often, Sometimes, Rarely, Never, I don't usually pass here.
language	string	en or tl.
submittedAt	timestamp	Submission time.

Implementation Notes

- A completed trip should always be saved to rides, even when deviationCount is 0.
- Only create deviation documents when actual route deviations are detected.
- If deviationCount is 0, post-trip questionnaire submission can mark the ride reviewed/complete without deviation questions.
- Use canonical English/internal answer values even when the rider views Tagalog labels.
- Current app does not yet persist real deviation events to Firebase. The rerouting detector updates the live route, but a separate persistence step is needed to create deviations/{deviationId}.
- Avoid fabricating route metadata. Store null when street, edge, image, or faster/slower calculation is unavailable.

Suggested Firebase Paths

```
users/{userId}
rides/{rideId}
rides/{rideId}/map/points
rides/{rideId}/annotations/{annotationId}
rides/{rideId}/postTripQuestionnaire/response
rides/{rideId}/deviations/{deviationId}
rides/{rideId}/deviations/{deviationId}/response/current
```